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About Me:

I am currently pursuing a Diploma in Data Science from the Indian Institute of Technology Madras (IIT Madras), with a strong focus on Machine Learning and Artificial Intelligence. This project provided a valuable opportunity to leverage my full-stack skills to develop a practical solution for a real-world urban problem.

Description

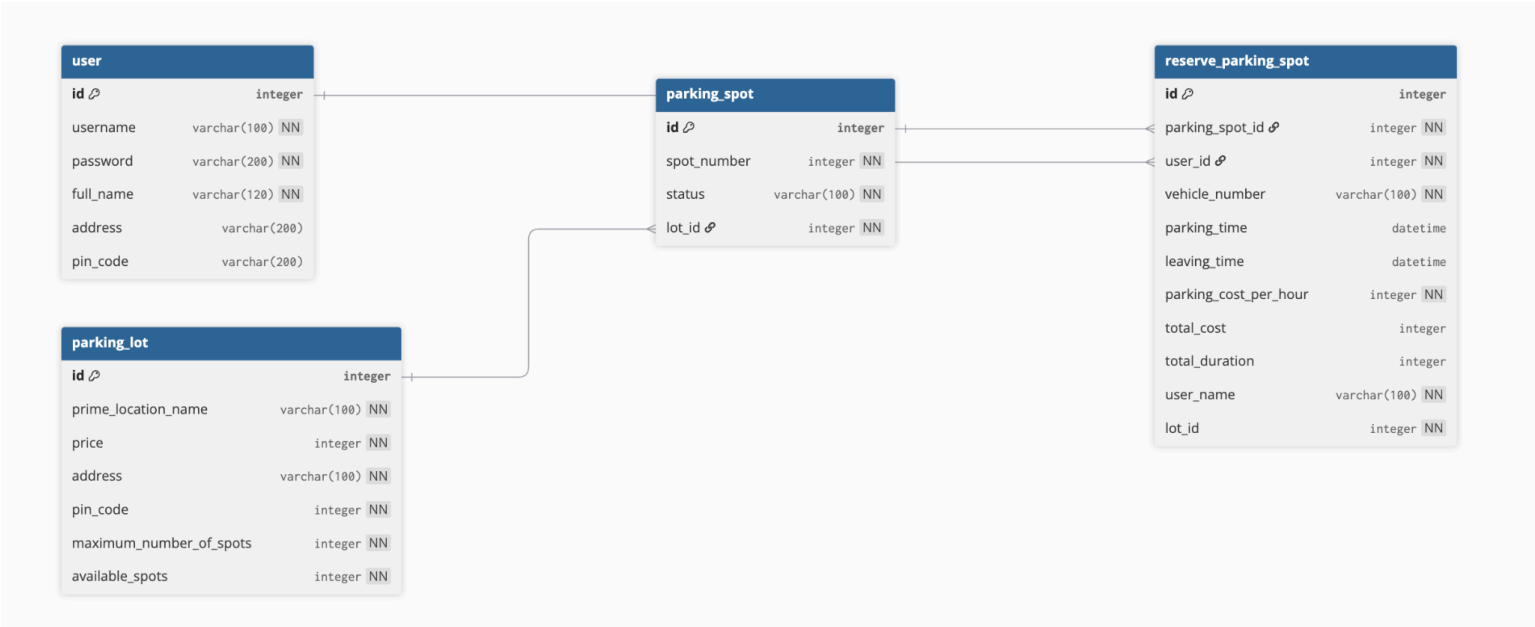
This project is a Flask-based Vehicle Parking Web Application designed with distinct functionalities for users and administrators.

1. **User Functionality:** Enables seamless, real-time searching, booking, and management of individual parking spots.
2. **Administrator Functionality:** Provides a comprehensive dashboard for overseeing parking lot status, monitoring spot usage, managing the user database, and analyzing key performance and revenue data through an interactive interface.

Technologies

This vehicle parking application was developed using a robust Python-based stack, chosen for its efficiency and ease of integration. The backend was engineered with the **Flask micro-framework**, utilizing **Flask-SQLAlchemy** as an ORM for seamless interaction with the **SQLite database**. Dynamic HTML pages were rendered using the **Jinja2** templating engine. For the frontend, **Bootstrap** was employed to create a fully responsive user interface. To provide

analytics, **Matplotlib** was integrated to generate visual charts of quiz results. Version control and collaboration were managed using **GitHub**.



File Structure

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- ├── app.py
- ├── config.py
- ├── models.py
- ├── Project Documentation.pdf
- ├── README.md
- ├── routes.py
- ├── static/
- └── templates/

The database is structured to efficiently manage a vehicle parking system. It separates data into distinct entities—Users, Parking Lots, and Parking Spots—and uses a central Reserve Parking Spots table to log transactions. This design ensures data integrity and allows for easy tracking of bookings, spot availability, and user activity.

- Users: Manages user profiles and credentials.

- Parking Lots: Stores details for each parking facility, including location, price, and capacity.
- Parking Spots: Tracks individual spot numbers and their availability status within a specific lot.
- Reserve Parking Spots: Acts as the booking ledger, linking users to spots and recording transaction details like time and cost.

Use of AI

I used a few AI tools to help me build this project. They were really useful for speeding up debugging and getting a hand with the Bootstrap layout, which made the whole process much smoother.

Click on this for project demo video → [Video link](#)